

## Department of Computer Applications

# FaceTally – Facial Attendance System

MAJOR PROJECT-II

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# INTRODUCTION

## Objective

- Automate student tracking using facial recognition
- Get rid of manual roll calls and reduce human errors
- Capture, identify and record attendance in real time

## The Problem

- Manual attendance is slow and easy to fake with proxies
- Paper records are messy, hard to search, and easy to lose
- No way to verify or monitor attendance in real time

## Our Solution

We built a Python-based system that uses FaceNet + SVM to identify students through a webcam in real time. It automatically records their attendance — including date, time, and status — into a MySQL database, with no manual input needed.



🎯 SDG 4 – Quality Education  
Supporting better classroom management through AI

# WHY FACIAL RECOGNITION?

## How We Stand Apart

### Contactless

No touching required — much more hygienic than fingerprint scanners.

### Proxy-Proof

You can't fake a face — no more sending a friend to mark your attendance.

### Real-Time

Powered by FaceNet deep learning — identification happens in milliseconds.

### No Duplicates

Once marked, the system blocks any repeat entry for the same student.

## Existing Systems & Their Limits

| System              | Limitation               |
|---------------------|--------------------------|
| Manual Roll Call    | Slow and error-prone     |
| Fingerprint Scanner | Requires physical touch  |
| RFID Cards          | Can be shared easily     |
| Basic CCTV          | No automatic recognition |

# TECHNOLOGY USED

| Category             | Technology                  |
|----------------------|-----------------------------|
| Programming Language | Python 3.10                 |
| Desktop GUI          | Tkinter                     |
| Computer Vision      | OpenCV                      |
| Deep Learning        | TensorFlow / Keras          |
| ML Classifier        | SVM via scikit-learn        |
| Database             | MySQL (via XAMPP / MariaDB) |

## ML Models Used



### FaceNet CNN

Converts each face into a unique 128-dimensional vector (embedding) that represents that person's facial features.



### SVM Classifier



A Support Vector Machine learns to distinguish between students based on their face embeddings. Fast and accurate.

*Confidence threshold: 60% or above for a valid match*

Python 3.10 • OpenCV • TensorFlow • MySQL

# SYSTEM ARCHITECTURE

| Module             | What It Does                                                   |
|--------------------|----------------------------------------------------------------|
| Admin Login        | Secure username and password authentication to protect access. |
| Student Management | Capture face photos via webcam and manage student records.     |
| Feature Extraction | FaceNet converts face images into 128-D vectors.               |
| Model Training     | An SVM classifier is trained on the extracted embeddings.      |
| Take Attendance    | Live webcam feed identifies students and logs to MySQL.        |
| Attendance Reports | View, search and filter all recorded attendance logs.          |

## Database Schema

### attendance

ID · Name · Department · Email · Date of Joining

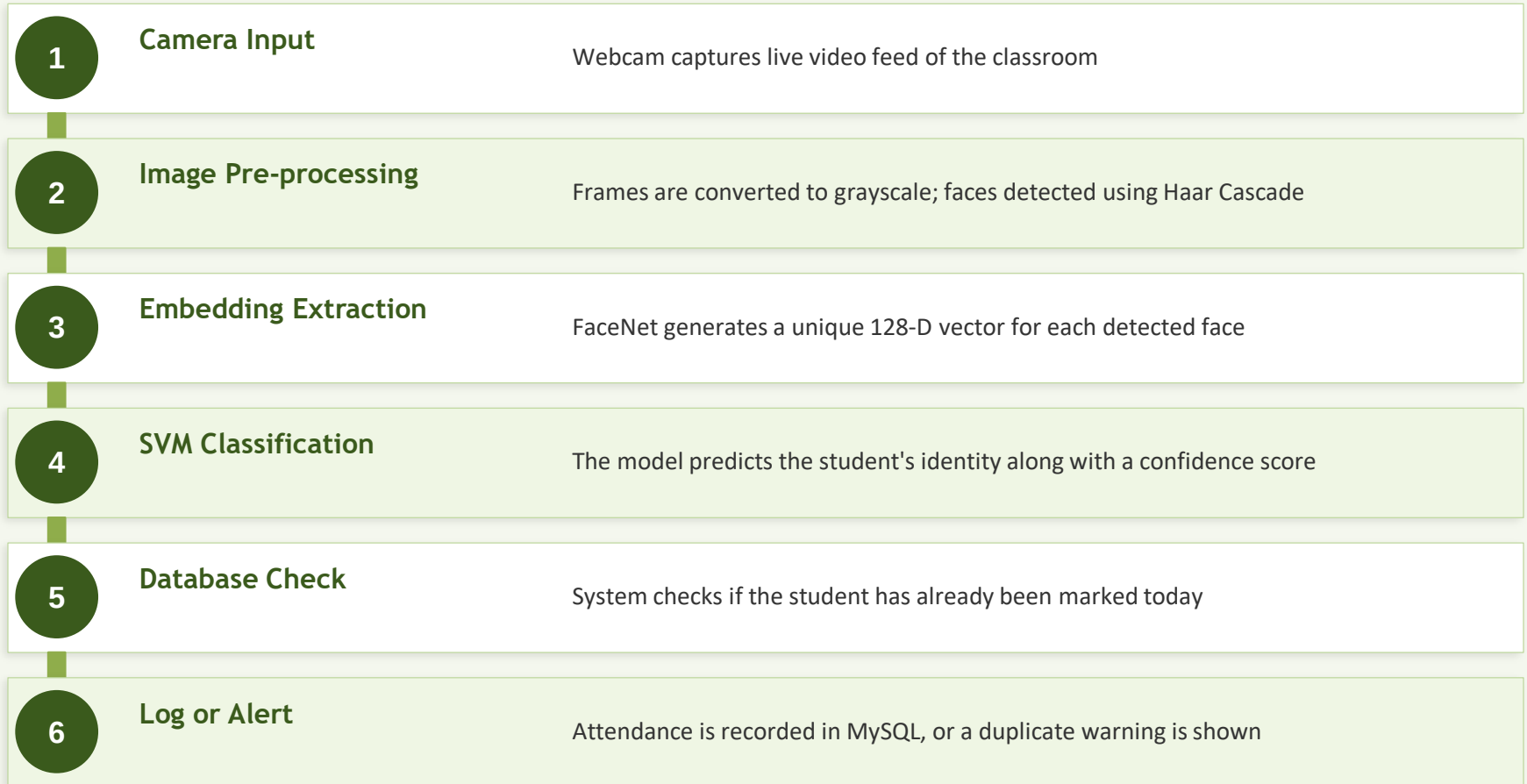
### report

Log ID · Name · Date · Time · Status

### login

Admin credentials (username + password)

# SYSTEM FLOWCHART



# WORKING METHODOLOGY

## STEP 01 · Registration

The admin enters student details, and the webcam captures 50 photos per student across different angles and lighting conditions to ensure strong training data.

## STEP 02 · Feature Extraction

Each photo is resized to 160×160 pixels. FaceNet processes them to generate unique 128-dimensional vectors, which are saved in a pickle file for later training.

## STEP 03 · Model Training

The face embeddings are labeled using LabelEncoder. A Linear SVM classifier is then trained on this data and saved as the recognition model.

## STEP 04 · Live Recognition

The webcam feed is analyzed frame by frame. If the model's confidence reaches 60% or above, the student's identity is confirmed and logged into MySQL automatically.

# Application Interface



## Login Page

Secure admin login with username & password



## Dashboard

Overview of students, attendance %, quick actions



## Face Recognition

Live webcam feed with real-time scanning



## Students List

Manage student records and face data



## Attendance Log

View and export all attendance records



## Reports

Daily/monthly reports with attendance charts



# FUTURE SCOPE



## Anti-Spoofing

Add liveness detection so the system can't be fooled by a photo or video of someone's face.



## Web Dashboard

Move from a desktop app to a Flask or Django web app so it can be accessed from anywhere.



## Cloud Deployment



Host the database and models in the cloud to support use across multiple campuses.



## Notifications

Send automatic email or SMS alerts to students or parents when attendance is marked.



## Live Webcam Support

Take facial recognition-based attendance through live webcam.



## Analytics

Generate visual heatmaps and trend charts to track attendance patterns over time.

## CONCLUSION

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*FaceTally shows that AI and computer vision can genuinely improve something as simple as taking attendance. It eliminates proxies, saves time, and keeps accurate records — all without anyone lifting a pen.*

**Thank You!**